

JIESHAN CHEN

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RESEARCH INTERESTS

AI4SE, SE4AI, Software Usability & Accessibility, Human-AI Collaboration, Responsible AI, UI Engineering, Dark Pattern Detection & Mitigation, Quality Assurance, Mobile Applications.

EDUCATION

Australian National University

August 2018 – December 2022

Ph.D. in Computer Science

Dissertation: Improving the Efficiency of Mobile User Interface Development through Semantic and Data-Driven Analyses

Supervisor: Zhenchang Xing

Sun Yat-Sen University

August 2014 – June 2018

Bachelor in Statistics, School of Mathematics

Honors Graduate Award (Top 10%)

Overall GPA: 3.8/4.0 (89/100)

WORKING EXPERIENCE

CSIRO, Australia

January 2022 – Present

Senior Research Scientist, Lead of UI Intelligence research

July 2025 - Present

Research Scientist, Lead of UI Intelligence research

January 2022 – July 2025

- Led research on responsible software engineering, LLM-agent systems, and human-AI collaboration, producing 17 research papers (14 CORE A*/Q1), in collaboration with the SE4AI team and 5 early-career, postdoctoral, PhD, and undergraduate researchers from the team and different universities nationally and internationally.
- Led and contributed to multiple responsible AI research and projects, advancing AI risk assessment and assurance frameworks, with outcomes integrated into NSW Government AI Assurance pilots and policy processes.
- Secured over 1 million AUD in competitive research funding (~261k) and project revenues.
- Initiated and co-organized international workshops (Harness4GenUI@ASE2026, SAML@ICSA2027, RAIE@ICSE2026, LAMPS@CCS2025) while representing the company in sponsoring and supporting broader academic collaborations (e.g., ICSE2023, WWW2025).
- Supported institutional governance and academic service through recruitment panels, internal award assessment committees, staff wellbeing and research-progress check-ins, and the stewardship of shared research resources.

Apple Inc., USA (Remotely from Australia due to Covid)

March 2021– September 2021

AI/ML Research Intern - Machine Learning + UI Understanding Team

- Project Topic: Enhancing the accessibility for screen readers and mobile users [C.4, P.1]
- Built a large-scale icon dataset (327 k samples) and developed an icon classification algorithm covering both common and long-tail icons, integrated into iOS Screen Recognition, and published a paper.

FUNDINGS

ACM SIGSOFT funding for Summer / Winter School

2025

TUM-IAS Dieter Schwarz Fellowship

2024-2027

HONORS & AWARDS

CSIRO SCS Biannual Awards – Early Career in Science Award

June 2024

CSIRO SCS Biannual Awards - Women in Science Career Award

June 2023

ACM SIGSOFT Distinguished Paper Award at ICSE 2020

July 2020

Google PhD Fellowship Nomination

2020

ANU HDR Fee Remission Merit Scholarship

2018-2022

ANU Ph.D. Scholarship (International) Full-Time

2018-2021

Honors Graduate Award at Sun Yat-Sen University

June 2018

PUBLICATIONS

CONFERENCE PUBLICATIONS - Research/Technical Track

[C.12] Zewei Shi, Ruoxi Sun, *Jieshan Chen*, Jiamou Sun, Jason Xue, Yansong Gao, Feng Liu, Xingliang Yuan. *50 Shades of Deceptive Patterns: A*

Last Updated in Oct 2025

Unified Taxonomy, Multimodal Detection, and Security Implications. Proceedings of the ACM Web Conference (**WWW2025**), Core Ranking A*, Acceptance rate: 19.8% (409/2062).

- [C.11] Ziqi Ding, Gelei Deng, Yi Liu, Junchen Ding, **Jieshan Chen**, Yulei Sui, Yuekang Li. *IllusionCAPTCHA: A CAPTCHA based on visual illusion*. Proceedings of the ACM Web Conference (**WWW2025**). Core Ranking A*, Acceptance rate: 19.8% (409/2062).
- [C.10] Mingyue Yuan, **Jieshan Chen**, Zhenchang Xing, Aaron Quigley, Yuyu Luo, Gelareh Mohammadi, Qinghua Lu, Liming Zhu. *DesignRepair: Dual-Stream Design Guideline-Aware Frontend Repair with Large Language Models*. Proceedings of the ACM/IEEE International Conference on Software Engineering (**ICSE2025**). Core Ranking A*, Acceptance Rate: 21.3% (245/1150).
- [C.9] Minh Duc Vu, Han Wang, Zhuang Li, **Jieshan Chen**, Shengdong Zhao, Zhenchang Xing, Chunyang Chen. *GPTVoiceTasker: Advancing Multi-step Mobile Task Efficiency Through Dynamic Interface Exploration and Learning*. Proceedings of the ACM Symposium on User Interface Software and Technology (**UIST2024**). Core Ranking A*, Acceptance Rate 24% (146/608).
- [C.8] Jiamou Sun, **Jieshan Chen**, Zhenchang Xing, Qinghua Lu, Xiwei Xu, Liming Zhu. *Where is it? Tracing the Vulnerability-Relevant Files from Vulnerability Reports*. Proceedings of the ACM/IEEE International Conference on Software Engineering (**ICSE2024**). Core Ranking A*, Acceptance Rate 22.3% (234/1051).
- [C.7] Shidong Pan, Dawen Zhang, Mark Staples, Zhenchang Xing, **Jieshan Chen**, Xiwei Xu, and Thong Hoang. *Is It a Trap? A Large-scale Empirical Study and Comprehensive Assessment of Online Automated Privacy Policy Generators for Mobile Apps*. Proceedings of the USENIX Security Symposium (**USENIX2024**). Core Ranking A*.
- [C.6] **Jieshan Chen**, Jiamou Sun, Sidong Feng, Zhenchang Xing, Qinghua Lu, Xiwei Xu, Chunyang Chen. *Unveiling the Tricks: Automated Detection of Dark Patterns in Mobile Applications*. Proceedings of the ACM Symposium on User Interface Software and Technology (**UIST 2023**). Core Ranking A*, Acceptance Rate 25% (121/483).
- [C.5] Qing Huang, Zhenyu Wan, Zhenchang Xing, Changjing Wang, **Jieshan Chen**, Xiwei Xu, Qinghua Lu. *Let's Chat to Find the APIs: Connecting Human, LLM and Knowledge Graph through AI Chain*. Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (**ASE2023**). Core Ranking A*, Acceptance Rate 20.3% (134/661).
- [C.4] **Jieshan Chen**, Amanda Swearngin, Jason Wu, Titus Barik, Jeffrey Nichols and Xiaoyi Zhang. *Towards Complete Icon Labeling in Mobile Applications*. Proceedings of the ACM Conference on Human Factors in Computing Systems (**CHI2022**). Core Ranking A*, Acceptance Rate 12.5% (324/2,597). [Available on iOS](#).
- [C.3] **Jieshan Chen**, Mulong Xie, Zhenchang Xing, Chunyang Chen, Xiwei Xu, Liming Zhu and Guoqiang Li. *Object Detection for Graphical User Interface: Old Fashioned or Deep Learning or a Combination?* Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE2020**). Core Ranking A*, Acceptance Rate 28% (101/360). [Used by Alibaba](#).
- [C.2] **Jieshan Chen**, Chunyang Chen, Zhenchang Xing, Xiwei Xu, Liming Zhu, Guoqiang Li, and Jinshui Wang. *Unblind Your Apps: Predicting Natural-Language Labels for Mobile GUI Components by Deep Learning*. Proceedings of the ACM/IEEE International Conference on Software Engineering (**ICSE2020**). Core Ranking A*, Cite 157, Acceptance Rate 20.9% (129/617). [ACM SIGSOFT Distinguished Paper Award](#)
- [C.1] Guibin Chen, Deheng Ye, Zhenchang Xing, **Jieshan Chen**, and Erik Cambria. *Ensemble Application of Convolutional and Recurrent Neural Networks for Multi-label Text Categorization*. Proceedings of the IEEE International Joint Conference on Neural Networks (**IJCNN2017**). Core Ranking A.

JOURNAL PUBLICATIONS

- [J.7*] Mingyue Yuan, **Jieshan Chen**, Dehai Zhao, Gelareh Mohammadi, Yoshifumi Kitamura, Aaron Quigley, Qinghua Lu, Zhenchang Xing. *Confidence vs. Competence: Misalignment in Judgment and Performance for Agentic Software Repair*. (**TOSEM 2026**). Just accepted (June)
- [J.6*] **Jieshan Chen**, Zhen Wang, Jiamou Sun, Wenbo Zou, Zhenchang Xing, Qinghua Lu, Qing Huang, Xiwei Xu. *From Exploration to Revelation: Detecting Dark Patterns in Mobile Apps*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**).
- [J.5] Mingyue Yuan, **Jieshan Chen**, Yongquan Hu, Sidong Feng, Mulong Xie, Gelareh Mohammadi, Zhenchang Xing, Aaron Quigley. *Towards Human-AI Synergy in UI Design: Supporting Iterative Generation with LLMs*. ACM Transactions on Computer-Human Interaction (**TOCHI2025**).
- [J.4] Qing Huang, Yanbang Sun, Zhenchang Xing, Yuanlong Cao, **Jieshan Chen**, Xiwei Xu, Huan Jin, Jiaxing Lu. *Let's Discover More API Relations: A Large Language Model-based AI Chain for Unsupervised API Relation Inference*. ACM Transactions on Software Engineering and Methodology (**TOSEM2024**).
- [J.3] Qing Huang, Zhiwen Luo, Zhenchang Xing, Jinshan Zeng, **Jieshan Chen**, Xiwei Xu, Yong Chen. *Revealing the Unseen: AI Chain on LLMs for Predicting Implicit Data Flows to Generate Data Flow Graphs in Dynamically-Typed Code*. ACM Transactions on Software Engineering and Methodology (**TOSEM2024**).
- [J.2] Yu Cheng, **Jieshan Chen**, Qing Huang, Zhenchang Xing, Xiwei Xu, Qinghua Lu. *Prompt Sapper: An LLM-Empowered Production Tool for Building AI Chains*. ACM Transactions on Software Engineering and Methodology (**TOSEM2024**).
- [J.1] **Jieshan Chen**, Chunyang Chen, Zhenchang Xing, Xin Xia, Liming Zhu, John Grundy, and Jinshui Wang. *Wireframe-based UI Design Search through Image Autoencoder*. ACM Transactions on Software Engineering and Methodology (**TOSEM2020**).

OTHER PUBLICATIONS – Short Paper/Tool Demo/Workshop

- [O.3] Sidong Feng, Mingyue Yuan, **Jieshan Chen**, Zhenchang Xing, Chunyang Chen. *Designing with Language: Wireframing UI Design Intent with Generative Large Language Models*. ACM CHI Conference on Human Factors in Computing Systems (CHI'24) - UI Workshop, **2024**.
- [O.2] Zewei Shi, Ruoxi Sun, **Jieshan Chen**, Jiamou Sun, Minhui Xue. *The Invisible Game on the Internet: A Case Study of Decoding Deceptive Patterns*. The Web Conference 2024 (WWW2024) - Short Papers Track, **2024**.
- [O.1] Mulong Xie, Sidong Feng, Zhenchang Xing, **Jieshan Chen**, Chunyang Chen. *UIED: A Hybrid Tool for GUI Element Detection*. Proceedings of the ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering – Companion (ESEC/FSE 2020 - Tool Demo), **2020**.

PREPRINT & UNDER REVIEW

- [P.6] Jingwen Zhou, **Jieshan Chen**, Qinghua Lu, Dehai Zhao, Liming Zhu. *SHIELDA: Structured Handling of Exceptions in LLM-Driven Agentic Workflows*. Under Review (TOSEM), **2025**.
- [P.5] Sai Zhang, Zhenchang Xing, **Jieshan Chen**, Dehai Zhao, Zizhong Zhu, Xiaowang Zhang, Zhiyong Feng, Xiaohong Li. *Think Like an Engineer: A Neuro-Symbolic Collaboration Agent for Generative Software Requirements Elicitation and Self-Review*. Under Review (TOSEM), **2025**.
- [P.4] Jingwen Zhou, Qinghua Lu, **Jieshan Chen**, Liming Zhu, Xiwei Xu, Zhenchang Xing, Stefan Harrer. *A Taxonomy of Architecture Options for Foundation Model-Based Agents: Analysis and Decision Model*. Under Review (TOSEM), **2025**.
- [P.3] Minh Duc Vu, **Jieshan Chen**, Zhenchang Xing, Qinghua Lu, Xiwei Xu, Qian Fu. *FactFlow: Automatic Fact Sheet Generation and Customization from Tabular Dataset via AI Chain Design & Implementation*. arXiv:2502.17909, 2024.
- [P.2] Jason Ching Yuen Siu, **Jieshan Chen**^{*}, Yujin Huang, Zhenchang Xing, Chunyang Chen. *Towards Real Smart Apps: Investigating Human-AI Interactions in Smartphone On-Device AI Apps*. arXiv preprint arXiv:2307.00756, 2023.
- [P.1] **Jieshan Chen**, Amanda Swearngin, Jason Wu, Titus Barik, Jeffrey Nichols and Xiaoyi Zhang. *Extracting Replayable Interactions from Videos of Mobile App Usage*. arXiv preprint arXiv:2207.04165, 2022.

Papers under conference review are not shown.

TEACHING

TUM Summer Term 26: Seminar on User Research Methods in Software Engineering

April 2026–July 2026

Lecturer & Course Convenor

Course Code: IN0014/IN2107/IN45137/INHN0015/ INHN4019); Class Size: 18

- Delivered the seminar remotely for the second time; Ongoing course.

TUM Winter Term 25|26: Seminar on User Research Methods in Software Engineering

October 2025– February 2026

Lecturer & Course Convenor

Course Code: IN0014/IN2107/IN45137/INHN0015/ INHN4019); Class Size: 11

- Independently designed the course materials and assignments from scratch
- Developed a cross-level seminar for undergraduate and graduate students from diverse disciplinary backgrounds, including Robotics, Business, Information Systems, and Computer Science.
- Delivered remotely through live online sessions.
- Teaching Evaluation: 100% student recommendation rate; 80% nominated for teaching award.**

MENTORING & ADVISING

Postdoctoral Researcher

- Jingwen Zhou [P.4, P.6] (Mar. 2024 – Present), CSIRO

PhD Students

- Zhicheng Bao (2026-Present), CSIRO & University of New South Wales
- Mingyue Yuan [O.3, O.4, C.10, J.5, J.7] (Mar. 2023 – Jun. 2026), CSIRO & University of New South Wales
- Zewei Shi [O.2, C.12] (2024 – 2025), CSIRO & University of Melbourne
- Sai Zhang [P.5] (2024 – 2025, Exchange Student), Tianjin University

Research Engineer

- Dustin Minh Duc Vu [C.9, P.3] (2023 – 2024), CSIRO

Undergraduate/Postgraduate Thesis students – Final year thesis project

- Erkin Sahin (2025-2026), Master's Student, Technical University of Munich.
- Yu Cheng [J.2] (2023), Undergraduate Student, Jiangxi Normal University.
- Jason Ching Yuen Siu [P.2] (2022), Undergraduate Student, Monash University
- Ziqi Zhao (2023), Master's Student, Monash University
- Xincheng Xu, Yukang Liu (2020-2021), Undergraduate Student, Australian National University
- Zhaowen Xu, Kexin Zhang, Mulong Xie [C.3, O.1] (2019-2020), Undergraduate Student, Australian National University
- Kai Xi (2018-2019), Undergraduate Student, Australian National University

Group Research Project Supervision

- Alex Zhou, Sam Howard, Nibert Bayer, Yicheng Peng, Prithviram Prabhuram, Chen Liu. (2023), Undergraduate Student, Monash FIT 4701

SERVICE

Chair

- The 6th International Workshop on Software Architecture and Machine Learning (SAML@ICSA2027)
- The 1st Workshop on Harness Engineering for Generative UI (Harness4GenUI@ASE2026)
- The International Conference on Software Architecture – Software Architecture Showcase Track (ICSA27-SA Showcase Track)
- The 4th International Workshop on Responsible AI Engineering (RAIE@ICSE2026)
- ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis (LAMPS@CCS2024, LAMPS@CCS2025)

Program Committee

- IEEE/ACM International Conference on Software Engineering (ICSE'25, ICSE'26, ICSE'27)

- IEEE/ACM International Conference on Automated Software Engineering (ASE'23, ASE'24 - Artifact Evaluation, ASE'25)
- ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'24)
- IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'24 ERA Track)
- ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023-SRC Track)
- ACM international joint conference on pervasive and ubiquitous computing (UbiComp/ISWC'23 Posters and Demos)
- International Conference on Software and System Processes (ICSSP'23)
- IEEE International Symposium on Software Reliability Engineering (ISSRE'22- Industry Track)
- ACM CHI Conference on Human Factors in Computing Systems (CHI'22-Computational UI Workshop)
- The Mining Software Repositories (MSR'22-Shadow PC, MSE'23-Junior PC)

Reviewer

- ACM Symposium on User Interface Software and Technology (UIST'22, UIST'24, UIST'25, UIST'26)
- ACM CHI Conference on Human Factors in Computing Systems (CHI'22, CHI'23, CHI'24, CHI'25, CHI'26)
- Transactions on Information Forensics & Security (IEEE-TIFS'26)
- IEEE Transactions on Software Engineering and Methodology (TOSEM'24, '25, '26)
- ACM Transaction on Computer-Human Interaction (TOCHI)
- ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
- ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW '24)
- IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR'23)
- The Mining Software Repositories (MSR'22, MSR'23)
- ACM international joint conference on pervasive and ubiquitous computing (UbiComp'22)
- International Journal of Human-Computer Interaction (IJHCI'22)
- IEEE International Symposium on Software Reliability Engineering (ISSRE'22)
- IEEE Transactions on Software Engineering (TSE'22, TSE '23)